EXPERIENCE · 1993 – 2026

Filip Truyens



More than thirty years in IT. I started my career in technical roles, then led programs in organizations where mistakes are expensive: banks, insurers, government. Today I advise on AI and, as a Forward Deployed Engineer, build it on site with the client, under my own banner: Fender ICT · AI Advisory.

33+	20	27
years in IT, since April 1993	employers and clients	years running my own company

All roles at a glance

PERIOD	DURATION	ORGANIZATION · ROLE
TODAY · 2026 – PRESENT		
Oct 2026 – present		**Fender ICT · AI Advisory** Founder · AI Advisor & Forward Deployed Engineer
PROGRAMS AND LEADERSHIP · 2010 – 2026		
Mar 2021 – Sep 2026	5 yrs 7 mos	**Smals** Program Manager · Assistant IT Director
Nov 2018 – Feb 2021	2 yrs 4 mos	**Burger Brands Belgium** Head of IT BeLux
May 2018 – Oct 2018	6 mos	**Belfius** Project Manager Office 365
Nov 2015 – Apr 2018	2 yrs 6 mos	**Getronics BeLux** Project Manager Managed Services
Oct 2014 – Oct 2015	1 yr 1 mo	**Deutsche Bank Belgium** Project Manager Application Services
Feb 2011 – Sep 2014	3 yrs 8 mos	**AXA Technology Services Belgium** Program Manager · Project Manager
Jan 2010 – Jan 2011	1 yr 1 mo	**De Persgroep Publishing** Team Lead · Project Manager · Infrastructure Architect
INFRASTRUCTURE AND ARCHITECTURE · 1993 – 2009		
Oct 2008 – Dec 2009	1 yr 3 mos	**Eandis** Infrastructure Architect
Jan 2005 – Sep 2008	3 yrs 9 mos	**Pioneer Europe** Senior Systems Engineer
Sep 2004 – Dec 2004	4 mos	**Agfa HealthCare** Systems Engineer, Integration Engineering
Sep 2002 – Aug 2004	2 yrs	**Belgian Post Group** Senior Systems Engineer

PERIOD	DURATION	ORGANIZATION · ROLE
Mar 2001 – Aug 2002	1 yr 6 mos	**Merck Sharp & Dohme** System Engineer, IS EMEA NT Computing
Dec 2000 – Feb 2001	3 mos	**Telenet** System Engineer
Nov 2000	1 mo	**CGU Benelux** System Engineer
Feb 2000 – Oct 2000	9 mos	**Distrirest Benelux** System Engineer
Jan 2000	1 mo	**UUNET** System Engineer
Mar 1998 – Dec 1999	1 yr 10 mos	**MSP bvba** System Engineer
Mar 1996 – Feb 1998	2 yrs	**CW Lease Belgium** NT System Administrator & AS/400 Operator
Aug 1995 – Nov 1995	4 mos	**MFS Communications** Data Analyst
Apr 1993 – Jul 1995	2 yrs 4 mos	**Zuidnatie** Novell System Administrator & Functional Analyst

Today

2026 – present

My own company, full time: advice on AI and co-creation as a Forward Deployed Engineer, with AI as my everyday tool.

Oct 2026 – present

Schilde

Own company

Fender ICT · AI Advisory

Founder · AI Advisor & Forward Deployed Engineer

My own company, founded in February 1999. It started as a management company and is now the base for AI Advisory: advice on AI, and co-creation as a Forward Deployed Engineer at the client.

WHAT I AM WORKING ON

- First client: Puur Reizen. As a Forward Deployed Engineer, set up a shared AI workspace in ChatGPT Business together with the team, with several projects and useful shared skills.
- Since late 2025 I have been building my own AI working environment: a growing library of skills, a memory system and AI agents I configure myself and use every day for research, software and documentation.
- For my clients, analysis and documentation work that used to take **weeks** now takes **days**.

Programs and leadership

2010 – 2026

Programs, projects and IT leadership in government, banking, insurance, IT services,

media and even retail.

Mar 2021 – Sep 2026
5 yrs 7 mos

Brussels

Public sector

Smals

Program Manager · Assistant IT Director

The shared IT provider for Belgian social security and the federal government. I ran programs that touched the whole organization and its member institutions, in a heavily regulated environment.

WHAT I BROUGHT

- Ran **three strategic programs at once**: public cloud, cybersecurity and emerging technologies, reporting to senior management through recurring steering committees.
- Turned regulation (NIS-2, CyberFundamentals, ISO 27001) and technical risk into priorities, budgets and roadmaps that management could decide on.
- In 2026, built **two working applications** with AI agents: a NIS-2 evidence factory and a capacity planner for some forty technical teams.

ROLES

Program Manager Post-Quantum Cryptography

Sep 2025 – Sep 2026 · 1 yr 1 mo

- Made senior management aware of what quantum computing means for today's encryption.
- Drew up a post-quantum readiness roadmap with Smals Research and took part in the EU-funded workshops.
- Worked out a crypto-agility approach with Enterprise Architecture, moving toward a single view of all hardware and software components (a global unified bill of materials).
- Wrote a vision paper on lifecycle management to reduce technical debt across infrastructure, platforms and applications.

Program Manager Public Cloud

Jun 2024 – Sep 2026 · 2 yrs 4 mos

- Set up the program: governance, roadmap, and the launch and follow-up of every project.
- Defined the public cloud strategy with C-level executives (Azure, AWS, Google Cloud).
- Brought procurement, internal security, finance, HR and the AI Competence Center together on public tenders through the EU framework contract OCRE-GÉANT, data sovereignty, FinOps, training and the first generative AI pilots.
- Coached technical project managers through studies, proofs of concept and pilots: landing zones, automation, governance and confidential computing.
- Helped set up the **Public Cloud Center of Excellence** for the public social security institutions (OISZ) and co-wrote its strategic principles.
- Worked with the federal public cloud task force on policies and technical policy packs, managed cloud providers and integrators, and ran info sessions for member institutions.

Technology: Azure, AWS, Google Cloud, Entra ID, PIM, FinOps, DevOps, CloudOps, CAF, CI/CD, confidential computing

Program Manager Security

Jan 2023 – Sep 2026 · 3 yrs 9 mos

- Led a self-assessment of the security posture with all stakeholders across Smals.
- Built a multi-year program to close the gaps, weighed on cost, risk, priority and feasibility.
- Oversaw the rollout of new security services (SOC, SIEM, CSIRT, vulnerability management, threat intelligence) and products (EDR/XDR, pentesting, forensics, honeypots) for Smals and its members.
- Paved the road to NIS-2 compliance in close contact with the CCB, with policies and procedures.
- In April 2026, built an automated evidence factory for NIS-2 and CyberFundamentals. It pulls configuration and control data from dozens of systems through their APIs and turns it into audit-ready reporting. **Weeks of manual work before every audit** became a repeatable run.

Technology: ISO 27001/2, NIS-2, CCB CyberFundamentals, NIST CSF, C2M2, OWASP SAMM, COBIT

Project Portfolio Manager

Jul 2022 – Sep 2026 · 4 yrs 3 mos

- Co-wrote the infrastructure roadmap in line with Smals' strategy, then steered its execution.
- Helped prioritize the Infrastructure & Support portfolio and set up recurring reporting.
- Matched people and profiles to the expected workload, and helped improve project methodology and capacity management.
- In July 2026, built a Resource Capacity Planner: one live view of operational load against project work for **some forty technical teams**, at three levels (team lead, division head, Office of the CTO). From data model to dashboards, running in production on Google Cloud. The tool replaced spreadsheets nobody fully trusted.

Technology: Sciforma, Confluence, Jira, Google Cloud

Transversal Project Manager

Mar 2021 – Jun 2023 · 2 yrs 4 mos

- Steered Smals' delivery and customer-facing teams toward one highly available, highly secure DevOps platform for several health and social care applications.
- Customers: RIZIV-INAMI (Recip-e, UHMEP, Vidis, SAMv2) and the Flemish Agency for Care and Health (Vitalink FHIR).
- Brought containers, CI/CD, privileged access management and secrets management together in one managed service.
- Owned it end to end: planning, communication, reporting to the customer and DPOs, technical implementation, support handover, cost calculation and invoicing.

Technology: Hyperconverged compute, VMware, Red Hat OpenShift, PostgreSQL, Ansible, CloudForms, Fluentd, Elasticsearch, AMQ, CyberArk PAM, CyberArk Conjur, Wallix PAM

Nov 2018 – Feb 2021
2 yrs 4 mos
Antwerp
Retail and franchising

Burger Brands Belgium

Head of IT BeLux

Operator and franchisor of three quick-service brands in Belgium and Luxembourg: Quick, Burger King and GoFish. More than 110 restaurants, company-owned and franchised, across five sales channels. I led IT with a team of eight and sat on the leadership team.

WHAT I BROUGHT

- **€266K** saved in 2019 on an annual IT operating budget of €1.5M.
- **15%** higher franchisee satisfaction with IT, as measured in a survey.
- Outsourced first-line support to partner NCR in Belgrade.
- Organized the IT team of eight into three cells: operations, support and projects.
- Delivered IT for **15 to 20** Burger King and GoFish openings and Quick conversions a year, across till, kiosk, delivery, drive-thru and click & collect.
- Led the rollout of ordering kiosks, the optimization of the TALAN ERP system and the move from the loyalty app to click & collect.
- Ran public and private cloud (Azure, Office 365) locally and for the group, with peers in Luxembourg, France, Italy, Germany, Poland and Switzerland, and handled IT requests from headquarters and the Burger King group for Southern and Eastern Europe.
- Prepared the CAPEX and OPEX budgets, renegotiated supplier contracts, supported the DPO on GDPR and presented IT plans and their cost to franchisees and the board.

Technology: Azure, Office 365, Power BI, QlikView, Jira

May 2018 – Oct 2018
6 mos
Brussels
Banking and insurance

Belfius

Project Manager Office 365

Belgian bank and insurer. I ran the risk and compliance stream of the strategic Office 365 program.

WHAT I BROUGHT

- Got the National Bank of Belgium, Microsoft and internal functions (legal, security, non-financial risk, compliance, data protection, contracts and audit) aligned.
- Took the program through the internal and regulatory advisory process on outsourcing: materiality assessment, Cloud Risk Assessment and DPIA.
- Laid out the path for deploying and managing Office 365 at Belfius.

Nov 2015 – Apr 2018
2 yrs 6 mos
Brussels
IT services

Getronics BeLux

Project Manager Managed Services

IT service provider in Belgium and Luxembourg. I led infrastructure projects for managed services customers, with teams in Belgium and India.

WHAT I BROUGHT

- Migrated **more than 500 Argenta branches** and several partners to a new DMZ, together with other suppliers. Any failure meant branch downtime. Delivered on time, to specification and within budget.
- Also moved an Argenta data center to Geleen (the Netherlands).

- Led an end-to-end insourcing for Argenx: infrastructure (vBlock, Office 365, SIP trunk, MPLS) and managed services (service desk, field services, monitoring).
- Led the LAN refresh with Cisco ISE for Record Bank, the WAN renewal for Bank Nagelmackers and a digital workspace for the World Steel Association.
- Headed the eight-month digital collaboration project for the City of Mechelen: Cisco telephony, contact center, Jabber and WebEx.
- Migrated Getronics' own monitoring from Smarts and NetIQ to ManageEngine, with a team at Getronics India.
- Guided internal process improvement with BPMN and helped the bid management team write proposals.

ROLES

Information Security Manager

Nov 2016 – Oct 2017 · 1 yr

Alongside my project management role.

- **ISO 27001** and **ISAE 3402** (type I and II) certification for Getronics Belgium and Luxembourg, achieved **in ten months**.
- Set up the ISMS: business case and scope, risk assessment and statement of applicability, policies and procedures, internal and external audits, management reviews.
- Reported directly to the CEO, CFO, COO and CISO, and helped prepare for GDPR.

Oct 2014 – Oct 2015
1 yr 1 mo
Brussels
Banking

Deutsche Bank Belgium

Project Manager Application Services

The Belgian arm of Deutsche Bank. I worked in the Online Banking domain.

WHAT I BROUGHT

- Led the release and environment management streams in Vittoria, the European Online Banking program of **over €15M**, built on ACI, Backbase and Alfresco. Also acted as product owner (Scrum).
- Steered several streams in the Essential Maintenance program (€4.4M), largely to remediate group audit findings.
- Renewed the technical stack of Online Banking and the Deutsche Bank Portal in Belgium, together with IBM.
- Secured the integrity of all source code developed in Belgium, with full traceability from development to production, in line with group SDLC standards and banking regulators' recommendations.

Technology: Subversion, Git, Maven, TeamCity, Nexus, uDeploy, Veracode, Apache, IIS, .NET

Feb 2011 – Sep 2014
3 yrs 8 mos
Brussels
Insurance

AXA Technology Services Belgium

Program Manager · Project Manager

The AXA group's IT provider, North and East Europe region, with data centers in Germany. I led a group-wide program and several infrastructure projects there.

WHAT I BROUGHT

- Led Nostradamus (RMPS2) for three years, a transformation program of **over 2,000 person-days** for AXA Group Solutions in France.

- Delivered a private cloud that became **the benchmark for all risk modelling applications** across the AXA group.
- Also led infrastructure projects for AXA Bank Belgium and for the PeopleSoft implementations of AXA Group Solutions and AXA Life Japan.

Name change: AXA Group Operations since 2019.

ROLES

Program Manager Risk Management Platform Solutions (Nostradamus)

Oct 2011 – Sep 2014 · 3 yrs

- Wrote the business case with three scenarios (private, hybrid and public cloud) and positioned AXA Tech in direct competition with Amazon AWS and Microsoft Azure, on price, service and expertise.
- Built the private cloud from March 2012, with a self-service portal and hourly, usage-based charge-back. Delivered on **11 July 2012** after four months of building: on time, on budget and to specification.
- Onboarded the first customer, AXA Hedging Services, in September 2012, followed by other risk modelling customers.
- Defined the operating model (incident, problem, change), with technical and billing reporting, and negotiated new SLAs.
- Steered the move from IBM to HP blades and added the private cloud offer to the service catalog.
- Helped shape the group strategy for private and public cloud, defended the investments before the investment committee and promoted the platform to AXA entities worldwide.

Technology: VMware, Citrix PVS, Citrix XenApp, MS HPC, Tibco DataSynapse, BMC Atrium Orchestrator, HP blades, EMC SAN, NetApp NAS, Prophet, MoSes, Algo Risk, ReMetrica

Project Manager AXA Bank Belgium

Apr 2013 – Jun 2014 · 1 yr 3 mos

- Within the regional Pillar savings program, led a data lifecycle management project (150 person-days): an archive service that automatically moves dormant data to low-cost storage.
- Infrastructure project manager for the confidential Bank of Tomorrow study (65 person-days): replacing some 50 mainframe applications with Sopra banking software (Thaler) on Linux and Oracle.
- Refreshed the disaster recovery NAS in the German data center (60 person-days).

Project Manager AXA Corporate Programs

Feb 2011 – Dec 2012 · 1 yr 11 mos

- Led the infrastructure component of AXA Life Japan's PeopleSoft HR implementation (205 person-days, April 2011 to March 2012).
- Led the infrastructure component of AXA Group Solutions' PeopleSoft Finance (213 person-days, July 2011 to December 2012), including the transition of some 50 file transfers for eight customers in Asia-Pacific.

Jan 2010 – Jan 2011
1 yr 1 mo
Brussels
Media

De Persgroep Publishing

Team Lead · Project Manager · Infrastructure Architect

Publisher of newspapers and magazines in Belgium and the Netherlands. After the takeover of PCM Uitgevers, the Dutch newspapers' IT had to move away from outsourcer Getronics.

WHAT I BROUGHT

- Led a team of **ten systems engineers** (Windows and Mac) and acted as interim IT Infrastructure Manager.
- Infrastructure architect for the business applications (publishing, marketing, advertising): target architecture, proofs of concept and capacity planning.
- Steered optimization and cost-saving tracks and improved incident, problem and change management.

Name change: DPG Media since 2019.

ROLES

Project Manager PCM Uitgevers (De Persgroep Nederland)

Mar 2010 – Jan 2011 · 11 mos

- Ran the insourcing program of **over 2,000 person-days** for the exit from Getronics, for Algemeen Dagblad, Trouw and de Volkskrant.
- **1,800 workstations** and the Windows back end replaced in four months, with a team of 25.
- Migrated **2,000 mailboxes**, calendars and public folders in two months.
- Separated DNS and DHCP, consolidated Selligent, and acted as the link between Dutch IT, Belgian IT and outsourcer Getronics.

Technology: Windows Server 2008, Windows 7, Active Directory, SCCM, VMware ESX, Citrix XenApp, Exchange, Quest Migration Manager

Infrastructure and architecture

1993 – 2009

More than sixteen years of building: from system administration and networks to infrastructure architecture, often in international environments. Self-employed from February 1999.

Oct 2008 – Dec 2009
1 yr 3 mos
Energy

Eandis

Infrastructure Architect

Electricity and gas distribution system operator in Flanders. I streamlined the development, validation and production environments.

WHAT I BROUGHT

- Designed a new VMware VI3 platform for all pre-production environments and guided the transition.
- Drew up the strategy and roadmap for rolling out virtualization across the organization.
- Defined a standard ICT service offering for virtualization with outsourcing partner Telindus/HP and got it approved by the CAB.
- Built the full cost picture, raised the purchase orders and secured internal approval.

- Presented the new environment to developers, project managers, service managers and architects.

Name change: Merged with Infrax into Fluvius in 2018.

Jan 2005 – Sep 2008
3 yrs 9 mos
Antwerp
Electronics

Pioneer Europe

Senior Systems Engineer

The European organization of the Japanese electronics maker, with its European data center in Antwerp.

WHAT I BROUGHT

- Designed and built the IT for the new **European call center** in Antwerp: Citrix with thin clients at the front, clustered Windows and SQL Server servers behind.
- Designed a new Citrix platform (Presentation Server 4.5) for Pioneer and its European subsidiaries and migrated the existing MetaFrame XP farm.
- Upgraded the VMware environment (five ESX servers on blades with SAN), consolidated Windows and Linux servers and converted physical servers to virtual.
- Chose Altiris after comparing it with SMS, LANDesk and ZENworks, then built desktop management, software distribution and patch management for all European offices.
- Built endpoint security with McAfee ePolicy Orchestrator and Check Point Integrity, and set up quota management for the file servers.
- Tested Office Communications Server 2007 and web meeting tools in a separate environment, without touching the existing Active Directory.

ROLES

Project Manager

Jan 2008 – Sep 2008 · 9 mos

- Functional and technical lead for the company-wide SharePoint 2007 implementation, using PRINCE2.
- Compared SharePoint with IBM Lotus Quickr, and SCOM with Nagios, NetIQ AppManager and GroundWork Monitor.

Sep 2004 – Dec 2004
4 mos
Ghent
Healthcare IT

Agfa HealthCare

Systems Engineer, Integration Engineering

Supplier of imaging and software for hospitals.

WHAT I BROUGHT

- Built a Windows Terminal Server load-test environment for radiology information systems used in hospitals worldwide: simulation scripts, load tests and monitoring with MOM 2005.
- Also load-tested the Mediweb web application and wrote up the results for marketing.

Sep 2002 – Aug 2004
2 yrs
Brussels
Postal services

Belgian Post Group

Senior Systems Engineer

Belgium's postal company.

WHAT I BROUGHT

- Helped build a Citrix MetaFrame XP farm on IBM blades for Georoute, a business-critical application.
- Extended NetIQ AppManager monitoring to the entire Windows server estate, and trained colleagues and operators to use it.
- Built a highly available project management solution on Project Server 2003, together with Microsoft's product team. Microsoft turned it into a customer case story.
- Automated patch detection and deployment on **300 servers** with PatchLink, under change management and SLAs.
- Introduced server virtualization with Virtual Server 2005, consolidated web servers and SQL Server databases, and hardened the web servers in the perimeter network.

Name change: bpost since 2010.

Mar 2001 – Aug 2002
1 yr 6 mos
Brussels
Pharma

Merck Sharp & Dohme

System Engineer, IS EMEA NT Computing

The pharmaceutical group's European IT coordination center in Brussels.

WHAT I BROUGHT

- Rolled out NetIQ AppManager on **220 servers** across EMEA, with policy-based setup and reporting.
- Installed 20 new servers in EMEA subsidiaries remotely and unattended, including data and printer migration, and reinstalled 60 domain controllers.
- Set up a system to review and align the security standards of the Windows NT servers.
- Designed time synchronization for all Windows servers and helped plan the migration to Windows 2000 with US colleagues.

Dec 2000 – Feb 2001
3 mos
Mechelen
Telecom

Telenet

System Engineer

Cable and telecom operator in Flanders.

WHAT I BROUGHT

- Ran the entire internal Windows and BackOffice environment with three colleagues, about **50 servers**.
- Designed and built the Windows 2000 Active Directory and merged the legacy domains into one AD domain.
- Set up a new file structure on a Compaq SAN with Windows 2000 clustering, and distributed software with SMS 2.0.

Nov 2000
1 mo
Brussels
Insurance

CGU Benelux

System Engineer

Insurer. A short assignment.

WHAT I BROUGHT

- Documented the entire network in a Lotus Notes knowledge base for IT support, alongside Windows NT administration and second-line support.

Name change: Merged into Winterthur-Europe Assurances in 2002, part of AXA Belgium since 2008.

Feb 2000 – Oct 2000
9 mos
Tubize
Logistics

Distrirest Benelux

System Engineer

Logistics company building new offices and warehouses. I ran IT and moved it at the same time.

WHAT I BROUGHT

- Moved all Windows and AS/400 systems to the new building and set up disaster recovery procedures.
- Rebuilt the Windows NT architecture: Token Ring to Ethernet migration, new DHCP, WINS and DNS, logon scripts, profiles and policies.
- Prepared the technical infrastructure for the new Logys warehouse software (Windows NT and Oracle).
- Introduced cloned desktop deployment, built an intranet for support and wrote procedures for ISO certification.

Name change: Acquired by the STEF group in 1998, now STEF.

Jan 2000
1 mo
Brussels
Internet and telecom

UUNET

System Engineer

Internet service provider. A short assignment.

WHAT I BROUGHT

- Developed a laptop cloning method, wrote standard installation procedures and supported users on Windows NT and Exchange.

Name change: Then part of MCI WorldCom, Verizon since 2006.

Mar 1998 – Dec 1999
1 yr 10 mos
Edegem
IT services

MSP bvba

System Engineer

IBM Business Partner and Microsoft Solution Provider. First as an employee, then from February 1999 self-employed through my own company. Assignments from one day to several months at customer sites.

WHAT I BROUGHT

- Got LeasePlan Belgium's entire IT ready for the year 2000: with an inventory and software distribution tool, some **10 servers and 350 workstations** were compliant on time and fully documented.
- Ran an unattended, scripted workstation migration from Windows 9x to NT 4.0 for Rabobank Belgium.

- Built a completely new network for the Belgian Transport Workers' Union (BTB), with an AS/400 as file, DHCP and DNS server for about 30 PCs.
- Built a knowledge base and helpdesk (McAfee Helpdesk) for the Algemene Centrale during its move to Windows NT.
- Installed Windows NT on AS/400 and ran IBM Network Station pilots for other IBM customers.

Name change: Acquired by Cegeka in 2008.

Mar 1996 – Feb 1998
2 yrs
Aartselaar
Car leasing

CW Lease Belgium

NT System Administrator & AS/400 Operator

Company car leasing firm that was introducing many new systems at the time.

WHAT I BROUGHT

- Ran the entire PC network: five Windows NT servers and about **a hundred PCs**, including the upgrade from NT 3.51 to NT 4.0.
- Migrated the last Novell servers to Windows NT and built fault tolerance for the business-critical servers.
- Set up SQL Server with replication to the AS/400 (Datamirror) and was the company's technical expert for it.
- Backed up the AS/400 operator for the three systems and connected the PCs through SNA Server.
- Handled mail and remote access, wrote ISO procedures, and provided first-line support and training.

Name change: Later ING Car Lease (2001), Alphabet since 2011.

Aug 1995 – Nov 1995
4 mos
Genval
Telecom

MFS Communications

Data Analyst

Telecom company with offices across Europe.

WHAT I BROUGHT

- Took part in the rollout of the Lawson Financials ERP across the European offices, handling the data analysis.

Name change: Acquired by WorldCom in 1996, Verizon since 2006.

Apr 1993 – Jul 1995
2 yrs 4 mos
Antwerp
Port logistics

Zuidnatie

Novell System Administrator & Functional Analyst

Cargo handling company in the port of Antwerp. My first job in IT.

WHAT I BROUGHT

- Built a new network with an external software partner: two Novell servers and **25 OS/2 workstations**.
- Co-developed Dynahouse, new software for cargo handling and storage: translated user needs, tested and debugged, and organized acceptance.
- Single point of contact for all IT, including first-line support, procedures and training.